

Benno Käch | Curriculum Vitae

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Professional Summary

Machine Learning Researcher with a strong background in Physics (ETH Zurich, DESY) specializing in generative modeling and foundation models. Experience in training large-scale models on terabyte-scale datasets, bridging the gap between theoretical physics and applied AI. Currently, leading the development of SOTA small-molecule foundation models, surpassing industry benchmarks. Eager to leverage deep technical expertise in generative models, designing non-standard architecture with problem-specific inductive biases and treating data scarcity with self-supervised pretraining, multitask learning and the use of synthetic data.

Experience

In Virtuo Laboratories

Lugano, Switzerland

Machine Learning Researcher

2025–Present

- **Lead Researcher** for a small-molecule foundation model trained on billions of compounds. Designed a novel reinforcement learning–based algorithm for oracle optimization.
- **Surpassed NVIDIA's GenMol** on all key benchmarks: de novo generation, fragmented constrained generation, practical molecular optimization benchmark, and lead optimization.
- Led high-impact projects including surrogate docking on ultra-large molecular databases and hit identification analyses.

Deutsches Elektronen-Synchrotron (DESY)

Hamburg, Germany

Researcher (PhD/Postdoc)

2021–2024

- Developed novel generative model architectures (Flows, Transformers) for high-energy physics simulations, significantly reducing computational costs.
- Analyzed **petabytes of data** recorded by the CMS detector, utilizing advanced statistical frameworks to compare observations with theoretical expectations.
- Published results at top-tier ML workshops (NeurIPS) and physics journals, focusing on point cloud generation and mean-field attention mechanisms.

ETH Zurich, IWF

Zurich, Switzerland

Research Assistant

2020–2021

- Implemented ML-based process automation for production machinery.
- Supervised two Master's theses on computer vision topics (head pose detection and emotion recognition).
- Successfully transferred academic findings to industry partners, deploying models on live production hardware.

Simpego

Zurich, Switzerland

Data Analyst

2017–2020

- Engineered and deployed a boosted decision tree–based pricing model in R, optimizing risk assessment.
- Developed real-time monitoring tools to detect market shifts using quantile-based sampling for mixed data types.
- Built an in-house web scraping infrastructure using Selenium and BeautifulSoup, eliminating dependency on outsourced data providers.

Education

University of Hamburg

Ph.D. in Particle Physics

Thesis: Generative Modelling in High-Energy Particle Physics

Magna Cum Laude

2021–2024

ETH Zurich

M.Sc. in Physics

2019–2020

ETH Zurich

B.Sc. in Physics

2015–2018

Selected First Authored Publications

ICLR 2026: *Refine Drugs, Don't Complete Them: Uniform-Source Discrete Flows for Fragment-Based Drug Discovery (Main Track)*

NeurIPS 2023: *Pay Attention to Mean-Fields for Point Cloud Generation (ML4PS Workshop)*

NeurIPS 2022: *Point Cloud Generation using Transformer Encoders and Normalizing Flows (ML4PS Workshop)*

Awards

2025: Breakthrough Prize in Fundamental Physics (as member of the CMS Collaboration)

Technical Skills

Programming: Python (Proficient), C++ (Intermediate), R, SQL

Deep Learning: PyTorch, PyTorch Lightning, Ray, Weights & Biases, Geometric Deep Learning

Cloud/HPC: AWS, SLURM, Multi-GPU/Multi-Node Training, Docker

Languages

German: Native

English: Fluent

French: Conversational

Italian: Basic